

SQL Analysis of Singer-Songwriter Sales by Media Type

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CNM Ingenuity: Data Science Bootcamp (Cohort 20)

Agenda

01

Overview

Project scope, general approach & restrictions

02

Methods

Qualitative: How many Singer-Songwriters are there?

03

Part 1

Qualitative: How many Singer-Songwriters are there?

04

Part 2

Quantitative: Which media types have the most sales?

05

Up Next

Combining qualitative & quantitative approaches to support musicians

Project Evolution

Lens: Musician

What is interesting to me about this data as a musician?

Tool: SQLite3

Improve familiarity with SQL syntax and method chaining.

Scope: Chinook Data

What would my non-data music friends want to know about this data?

1: Singer-Songwriters

How many singer-songwriters are there?

2: Media Sales Trends

Which media type has the most invoice sales?

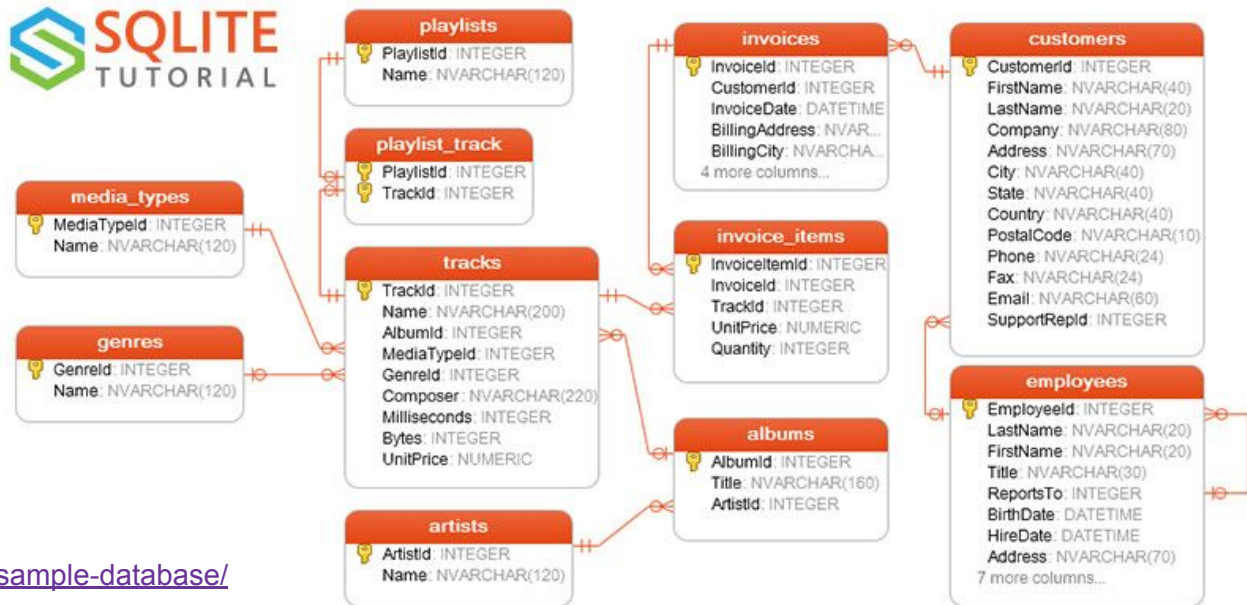
Long-term project

Is there a relationship between singer-songwriters and digital sales in Chinook data?

Methods

1. Confirm “source of truth” data
2. Lead with curiosity
3. Strict SQL Queries (per project guidelines)

Source: <https://www.sqlitetutorial.net/sqlite-sample-database/>



SELECT

LIMIT

DISTINCT

BETWEEN

GROUP BY

ORDER BY

MAX

MIN

WHERE

COUNT

SUM

AVG

AND

pandas

sqlite3

plotly

google.colab

ERD

Musician Persona

Initial Data Analysis

CTE

Part 1:

Singer-Songwriters

- Some artists are listed as composers, but not all composers are listed as artists.
- **275** unique Name values in artists
- **2,525** unique Composer values in tracks
- **402** returned singer-songwriters
 - Are these unique?
 - Further data cleaning required.

SELECT

FROM

WHERE

LIMIT

ORDER BY

DISTINCT

COUNT

INNER JOIN

Below: SQL query for unique composers & problematic output

```
2 SELECT DISTINCT COUNT(composer)
3 FROM tracks
4 ;
```

COUNT(composer)

2525

Wolfgang Amadeus Mozart
Wright, Waters
alan bergman/marilyn bergman/peggy lipton jones/quincy jones
antonio carlos jobim/norman gimbel/vinicius de Moraes
bart howard
bert brecht/kurt weill/marc blitzstein
berthold kaempfert/charles singleton/eddie snyder

Below: CTE method to return singer-songwriters

```
2 WITH legends AS(
3   SELECT *
4   FROM artists, tracks
5   WHERE artists.name = tracks.composer
6 )
7 SELECT COUNT(*) AS total_rows
8 FROM legends
9 ;
```

total_rows

402

Part 2: Sales by Media Type

- 5 unique media types, distinguished by “protected” status
- **MPEG audio files** (type “1”), are the most popular, suggesting Chinook represents a *digital* media store
 - **Total Sales** by Media Type support this hypothesis
- Discrepancy in total **UnitPrice** from **invoice_items** (~\$2.3k) vs. **tracks** (~3.7k)
 - Further investigation needed

SELECT

MAX

MIN

AVG

FROM

WHERE

BETWEEN

AND

ORDER BY

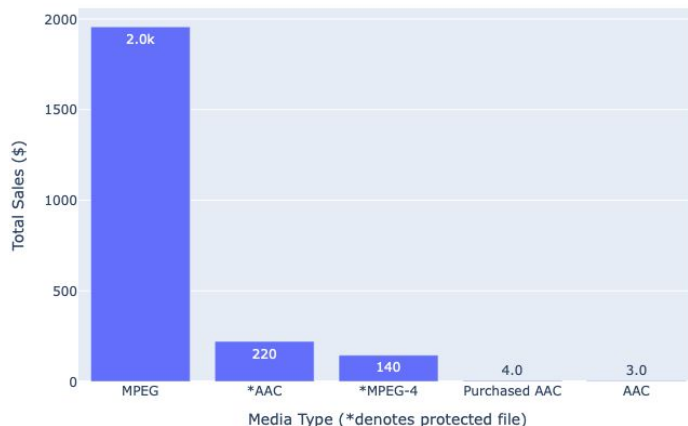
Below: Media types by track count, name, and sales, including the least popular tracks by media type (“4”)

MediaTypeId	track_count	Name
1	3034	MPEG audio file
2	237	Protected AAC audio file
3	214	Protected MPEG-4 video file
5	11	Purchased AAC audio file
4	7	AAC audio file

Name

War Pigs
Symphony No. 104 in D Major "London": IV. Finale: Spiritoso
SCRIABIN: Prelude in B Major, Op. 11, No. 11
Prometheus Overture, Op. 43
Sonata for Solo Violin: IV: Presto
Étude 1, In C Major – Preludio (Presto) – Liszt
Concerto for Violin, Strings and Continuo in G Major, Op. 3,

Chinook Total Sales by Media Type



Up Next

[View GitHub](#)

01

EDA

Clean qualitative data and investigate quantitative discrepancies

02

CTE

More advanced queries with complicated relationships (removed by 1)

03

Model

Hypothesis: Digital sales separate performers from songwriters, reducing singer-songwriters.

Thank you! Questions?